Algorithms

- Running Time
  - Test after 2 to 3 weeks of these slides
  - explicit formulas - $T(n) = 2^n + n - 1$

    $T(8) = 263$
  - recursive formula - $a_1 = 0,\ a_2 = 1,\ a_n = a_{n-1} + a_{n-2} \quad n > 1$
  - $T(n) = T(n-1) + 2$

    $T(n-1) = (T(n-2) + 2) + 2$

    $T(n-2) = (T(n-3) + 2 + 2 + 2)$

    $T(1) + \sum_{i=1}^{n-1} 2$

  - $T(1) = 1$

    $T(2) = 1 + 2$

    $T(3) = 1 + 2 + 2$

    $T(n) = 1 + \overbrace{2 + 2 + 2}^{n-1}$

    $T(n) = 1 + 2(n-1)$

    $T(n) = 2n - 1$

  - $T(n) = T(\frac{n}{2}) + 3$

    $T(\frac{n}{2}) = T(\frac{n}{4}) + 3 + 3$

    $T(\frac{n}{4}) = T(\frac{n}{8}) + 3 + 3 + 3$

    $= T(1) + 3(\lg n - 1)$

    $= 3 \lg n$

    $O(\lg n)$